Day 1: Physics Model and First Dataset Generation

Hydrogen-Induced Defect Transport Project

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Abstract

This document provides a detailed Day 1 plan for the project *Hydrogen-Induced Defect Transport*. The goal is to create a simple one-dimensional tight-binding lattice model, introduce hydrogen-induced defects as local perturbations, compute the energy spectrum, evaluate localization using the inverse participation ratio, define a simple transport-related descriptor, and generate the first machine-learning-ready dataset. The document is written in an Overleaf-ready format and includes scientific explanations, Python code, new terminology, and references.

English Correction

The original request can be written more naturally as:

> *"Now, please create all of this in a detailed Overleaf-ready text file."*

1 Week 1 Context

The general goal of Week 1 is to build the first physics-based simulation pipeline. By the end of Week 1, the project should include:

- a simple lattice model,
- hydrogen defect configurations,
- energy spectrum calculation,
- localization index calculation,
- transport-related descriptor calculation,
- the first dataset for later machine learning and quantum machine learning.

The scientific direction of the project can be summarized as:

Hydrogen defects → lattice Hamiltonian → energy spectrum → localization → transport descriptor → ML/QML dataset.

2 Day 1 Goal

The realistic goal for Day 1 is to complete the first notebook:

`notebooks/01_lattice_model.ipynb`

By the end of Day 1, we should have:

1. created the project folder,
2. created the first notebook,
3. written the scientific project paragraph,
4. built a clean one-dimensional lattice Hamiltonian,
5. added simple hydrogen-induced defects,
6. computed eigenvalues and eigenvectors,
7. computed the inverse participation ratio,
8. defined a simple transport descriptor,
9. generated and saved the first small dataset.

3 Project Folder Structure

Create the following folder structure:

```
hydrogen_defect_transport/
|
|-- data/
|-- notebooks/
|-- src/
|-- results/
|-- figures/
|-- manuscript/
|-- README.md
```

Folder Meanings

Folder	Purpose
`data/`	Stores generated datasets.
`notebooks/`	Stores Jupyter notebooks for exploration and explanation.
`src/`	Stores reusable Python functions later in the project.
`results/`	Stores numerical outputs and model results.
`figures/`	Stores plots for reports, presentations, and manuscripts.
`manuscript/`	Stores report or paper drafts.
`README.md`	Explains the project, installation, and usage.

4 Step 1: Create the Project Folder

4.1 Command for macOS or Linux

```
mkdir hydrogen_defect_transport
cd hydrogen_defect_transport

mkdir data notebooks src results figures manuscript
touch README.md
```

4.2 Command for Windows PowerShell

```
mkdir hydrogen_defect_transport
cd hydrogen_defect_transport

mkdir data, notebooks, src, results, figures, manuscript
New-Item README.md
```

4.3 Why This Step Matters

A clean folder structure is essential for reproducible scientific research. It separates code, data, results, figures, and manuscript writing. This will also make the project suitable for GitHub and future collaboration.

4.4 New Terms

Term	Explanation
Project structure	Organized folder system for a research project.
Reproducibility	The ability to run the same analysis again and obtain the same results.
README	A file explaining what the project is, how to install it, and how to run it.
Repository	A project folder managed by Git or hosted on GitHub.

5 Step 2: Create the Python Environment

5.1 Create a Conda Environment

```
conda create -n hydrogen_defect python=3.11
conda activate hydrogen_defect
```

5.2 Install Required Packages

```
pip install numpy pandas scipy matplotlib scikit-learn xgboost optuna
   jupyter
pip install qiskit qiskit-aer qiskit-ibm-runtime
```

5.3 Start Jupyter Notebook

```
jupyter notebook
```

Then create:

`notebooks/01_lattice_model.ipynb`

5.4 Why This Step Matters

A separate environment prevents dependency conflicts. For example, this project may require Qiskit and scientific packages that may not be compatible with older environments used in other projects.

5.5 Main Packages

Package	**Use**
`numpy`	Matrix construction and numerical operations.
`pandas`	Dataset table creation and CSV saving.
`scipy`	Eigenvalue and eigenvector calculations.
`matplotlib`	Scientific plotting.
`scikit-learn`	Later classical machine learning models.
`xgboost`	Later advanced machine learning baseline.
`optuna`	Later hyperparameter optimization.
`qiskit`	Later quantum circuits and quantum kernels.
`qiskit-aer`	Later quantum circuit simulation.
`qiskit-ibm-runtime`	Later IBM Quantum backend connection.

5.6 New Terms

Term	**Explanation**
Environment	Isolated Python workspace.
Dependency	A package required by the project.
Jupyter notebook	Interactive document containing code, equations, plots, and explanation.
Kernel	Python process running inside a notebook.

6 Step 3: Write the Project Paragraph

In the first Markdown cell of the notebook, write:

```
# 01 -- Lattice Hamiltonian Model for Hydrogen-Induced Defects

## Project definition

We investigate hydrogen-induced defects as local perturbations in a model
    lattice Hamiltonian and study their effect on electronic
   localization and transport-related descriptors. In this first
   notebook, we construct a simple one-dimensional tight-binding lattice
   , introduce hydrogen-like defects as local on-site energy
   perturbations, compute the energy spectrum, evaluate localization
```

```
using the inverse participation ratio, and generate the first physics
-informed dataset for later machine learning and quantum machine
learning analysis.
```

6.1 Scientific Meaning

The project paragraph defines the scientific story:

Hydrogen-induced defect $\to$ local perturbation $\to$ modified Hamiltonian $\to$ modified spectrum $\to$ modified localization $\to$ modified transport descriptor.

This gives a clear and publishable direction for the notebook.

6.2 New Terms

Term	Explanation
Hydrogen-induced defect	A defect caused or represented by the presence of hydrogen in a material.
Local perturbation	A change applied only to selected lattice sites.
Hamiltonian	Mathematical operator or matrix describing the energy of a physical system.
Energy spectrum	The set of allowed energy values of the system.
Localization	Concentration of a wavefunction in a limited spatial region.
Transport descriptor	Numerical quantity related to how easily an electron may move through the system.

7 Step 4: Import Libraries

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from scipy.linalg import eigh
```

7.1 Physics Meaning

We will solve the eigenvalue problem:

$$H\psi_n = E_n\psi_n, \tag{1}$$

where:

- H is the Hamiltonian matrix,
- E_n is the nth energy eigenvalue,
- ψ_n is the corresponding eigenvector or wavefunction.

7.2 Why Use eigh?

The Hamiltonian in quantum mechanics must be Hermitian. For real-valued tight-binding Hamiltonians, this means the matrix is symmetric:

$$H = H^T. \tag{2}$$

The function `scipy.linalg.eigh` is designed for real symmetric or complex Hermitian eigenvalue problems.

7.3 New Terms

Term	Explanation
Matrix	Rectangular array of numbers.
Eigenvalue	Special value representing an allowed energy in this context.
Eigenvector	State or wavefunction associated with an eigenvalue.
Hermitian matrix	Matrix whose eigenvalues are real; quantum observables are represented by Hermitian operators.

8 Step 5: Define Physical Parameters

```
N = 40                  # number of lattice sites
t = 1.0                 # hopping strength
epsilon_0 = 0.0         # base on-site energy
defect_strength = 2.0   # hydrogen-induced local perturbation
num_defects = 3         # number of hydrogen defects

np.random.seed(42)
```

8.1 Physics Meaning

We start with a one-dimensional chain:

$$\text{site } 0 - \text{site } 1 - \text{site } 2 - \text{site } 3 - \cdots - \text{site } 39$$

Each site represents a simplified atomic position. The electron can move from one site to the next. This movement is controlled by the hopping parameter t.

8.2 Clean Tight-Binding Hamiltonian

The clean one-dimensional nearest-neighbor tight-binding Hamiltonian is:

$$H = \sum_i \epsilon_0 |i\rangle\langle i| - t \sum_{\langle i,j\rangle} \left(|i\rangle\langle j| + |j\rangle\langle i|\right). \tag{3}$$

Here:

- ϵ_0 is the base on-site energy,
- t is the hopping amplitude,
- $\langle i, j\rangle$ means nearest-neighbor pairs,
- $|i\rangle$ represents the state localized at site i.

In matrix form, the Hamiltonian begins as:

$$H = \begin{bmatrix} \epsilon_0 & -t & 0 & 0 & \cdots \\ -t & \epsilon_0 & -t & 0 & \cdots \\ 0 & -t & \epsilon_0 & -t & \cdots \\ 0 & 0 & -t & \epsilon_0 & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots \end{bmatrix}. \tag{4}$$

8.3 New Terms

Term	Explanation
Lattice site	A discrete position where the electron can exist.
Hopping	Movement of an electron between neighboring sites.
On-site energy	Energy associated with an electron staying at a site.
Nearest neighbor	Directly connected neighboring site.
Tight-binding model	Simplified electronic model where electrons hop between lattice sites.

9 Step 6: Build the Clean 1D Hamiltonian

```python
def build_clean_hamiltonian(N, t=1.0, epsilon_0=0.0):
    """
    Build a clean 1D nearest-neighbor tight-binding Hamiltonian.

    Parameters
    ----------
    N : int
        Number of lattice sites.
    t : float
        Hopping strength between nearest neighbors.
    epsilon_0 : float
        Base on-site energy.

    Returns
    -------
    H : np.ndarray
        NxN Hamiltonian matrix.
    """
    H = np.zeros((N, N))

    # On-site energies
    np.fill_diagonal(H, epsilon_0)

    # Nearest-neighbor hopping
    for i in range(N - 1):
        H[i, i + 1] = -t
        H[i + 1, i] = -t

    return H
```

Run:

```python
H_clean = build_clean_hamiltonian(N, t, epsilon_0)

print(H_clean[:6, :6])
```

Expected output:

```
[[ 0. -1.  0.  0.  0.  0.]
 [-1.  0. -1.  0.  0.  0.]
 [ 0. -1.  0. -1.  0.  0.]
 [ 0.  0. -1.  0. -1.  0.]
 [ 0.  0.  0. -1.  0. -1.]
 [ 0.  0.  0.  0. -1.  0.]]
```

9.1 Physics Meaning

The diagonal entries represent on-site energies. The off-diagonal $-t$ entries represent hopping between neighboring sites.

9.2 New Terms

Term	Explanation
Diagonal term	Matrix element describing the energy of a site itself.
Off-diagonal term	Matrix element describing coupling between different sites.
Coupling	Interaction or connection between two sites or states.
Sparse structure	Matrix structure where most elements are zero.

10 Step 7: Add Hydrogen-Induced Defects

```python
def add_hydrogen_defects(H, num_defects=3, defect_strength=2.0, seed=None
    ):
    """
    Add hydrogen-like defects as local on-site energy perturbations.

    Parameters
    ----------
    H : np.ndarray
        Clean Hamiltonian.
    num_defects : int
        Number of defect sites.
    defect_strength : float
        Local perturbation strength.
    seed : int or None
        Random seed for reproducibility.

    Returns
    -------
    H_def : np.ndarray
        Defected Hamiltonian.
    defect_sites : np.ndarray
        Indices of defect sites.
    """
    if seed is not None:
        np.random.seed(seed)

    H_def = H.copy()
    N = H.shape[0]

    defect_sites = np.random.choice(N, size=num_defects, replace=False)
```

```
    for site in defect_sites:
        H_def[site, site] += defect_strength

    return H_def, defect_sites
```

Run:

```
H_def, defect_sites = add_hydrogen_defects(
    H_clean,
    num_defects=num_defects,
    defect_strength=defect_strength,
    seed=42
)

print("Defect sites:", defect_sites)
print(H_def[:6, :6])
```

10.1 Physics Meaning

A hydrogen-induced defect is represented as a local change in the on-site energy:

$$\epsilon_i \to \epsilon_i + V_H, \tag{5}$$

where V_H is the defect strength.

This is not yet a full atomistic hydrogen model. It is a simplified physics-inspired toy model. Later, the model can be improved by using two-dimensional lattices, realistic disorder distributions, DFT-based parameters, or transport calculations with leads.

10.2 New Terms

Term	**Explanation**
Defect site	Lattice position affected by a hydrogen-like perturbation.
Perturbation strength	Magnitude of the local energy change.
Disorder	Randomness or irregularity in the lattice.
Toy model	Simplified model used to understand the core physics before using realistic models.

11 Step 8: Compute Eigenvalues and Eigenvectors

```
eigenvalues, eigenvectors = eigh(H_def)

print("Number of eigenvalues:", len(eigenvalues))
print("First 5 eigenvalues:")
print(eigenvalues[:5])

print("Eigenvector matrix shape:", eigenvectors.shape)
```

11.1 Physics Meaning

We solve the eigenvalue problem:

$$H\psi_n = E_n\psi_n. \tag{6}$$

Each eigenvalue E_n is one allowed energy level. Each eigenvector ψ_n describes the spatial structure of the electron state.

In Python:

```
psi_0 = eigenvectors[:, 0]
```

This gives the wavefunction corresponding to the lowest energy eigenvalue.

11.2 New Terms

Term	Explanation
Energy eigenvalue	Allowed energy of the system.
Eigenstate	Quantum state corresponding to a specific energy.
Wavefunction	Amplitude distribution over lattice sites.
Ground state	Lowest-energy state.
Excited state	Higher-energy state.

12 Step 9: Plot the Energy Spectrum

```
plt.figure(figsize=(7, 4))
plt.plot(eigenvalues, marker="o", linestyle="none")
plt.xlabel("Eigenstate index")
plt.ylabel("Energy")
plt.title("Energy spectrum with hydrogen-induced defects")
plt.grid(True)
plt.show()
```

12.1 Physics Meaning

The energy spectrum shows the allowed energy levels of the defected lattice. Later, this can be compared with the spectrum of the clean lattice.

A useful comparison is:

$$\text{clean lattice spectrum} \quad \text{versus} \quad \text{defected lattice spectrum.}$$

Hydrogen defects may shift energy levels or create defect-sensitive states.

12.2 New Terms

Term	Explanation
Energy spectrum	Full list of energy eigenvalues.
Spectral shift	Change in energy levels caused by defects.
Defect state	State strongly influenced by a defect.
Band gap	Energy difference between selected energy levels.

13 Step 10: Compute Localization Using IPR

```python
def compute_ipr(eigenvectors):
    """
    Compute inverse participation ratio for each eigenstate.

    IPR_n = sum_i |psi_n(i)|^4

    Parameters
    ----------
    eigenvectors : np.ndarray
        Matrix whose columns are eigenvectors.

    Returns
    -------
    ipr : np.ndarray
        IPR value for each eigenstate.
    """
    ipr = np.sum(np.abs(eigenvectors) ** 4, axis=0)
    return ipr
```

Run:

```python
ipr_values = compute_ipr(eigenvectors)

print("Mean IPR:", np.mean(ipr_values))
print("Max IPR:", np.max(ipr_values))
print("Min IPR:", np.min(ipr_values))
```

13.1 Physics Meaning

The inverse participation ratio is defined as:

$$IPR_n = \sum_i |\psi_n(i)|^4. \tag{7}$$

If a wavefunction is spread over many sites, the IPR is small. If a wavefunction is concentrated on only a few sites, the IPR is large.

For a fully extended state over N sites:

$$|\psi(i)|^2 \approx \frac{1}{N}. \tag{8}$$

Then:

$$IPR \approx N \left(\frac{1}{N}\right)^2 = \frac{1}{N}. \tag{9}$$

For $N = 40$:

$$IPR \approx \frac{1}{40} = 0.025. \tag{10}$$

A strongly localized state will have a much larger IPR.

13.2 New Terms

Term	Explanation
IPR	Inverse participation ratio; a localization measure.

Extended state	Wavefunction spread over many sites.
Localized state	Wavefunction concentrated in a small spatial region.
Probability amplitude	Value of the wavefunction at a site.
Localization index	Numerical quantity measuring localization strength.

14 Step 11: Plot IPR Values

```
plt.figure(figsize=(7, 4))
plt.plot(ipr_values, marker="o")
plt.xlabel("Eigenstate index")
plt.ylabel("IPR")
plt.title("Localization index of eigenstates")
plt.grid(True)
plt.show()
```

14.1 Physics Meaning

This plot shows which eigenstates are more localized. Large peaks in the IPR plot may indicate defect-sensitive states.

14.2 Suggested Notebook Comment

> The IPR plot shows how localized each eigenstate is. Eigenstates with larger IPR values are more spatially concentrated, while smaller IPR values indicate more extended states. In this simplified model, hydrogen-induced defects are expected to increase localization for some states by modifying local on-site energies.

14.3 New Terms

Term	Explanation
IPR peak	Eigenstate with stronger localization.
Spatial concentration	Wavefunction mostly located near specific sites.
Defect-sensitive state	Eigenstate strongly affected by defects.

15 Step 12: Visualize One Eigenstate

```
state_index = np.argmax(ipr_values)
psi = eigenvectors[:, state_index]

plt.figure(figsize=(7, 4))
plt.plot(np.abs(psi) ** 2, marker="o")
plt.xlabel("Lattice site")
plt.ylabel(r"$|\psi(i)|^2$")
plt.title(f"Most localized eigenstate, index = {state_index}")
plt.grid(True)
plt.show()

print("Most localized state index:", state_index)
print("IPR:", ipr_values[state_index])
print("Defect sites:", defect_sites)
```

15.1 Physics Meaning

This plot shows where the most localized wavefunction lives on the lattice. If the largest probability density appears near defect sites, this suggests that the defect is influencing localization.

15.2 New Terms

Term	Explanation
Probability density	$\lvert\psi(i)\rvert^2$, probability of finding the electron at site i.
Most localized state	Eigenstate with the largest IPR.
Wavefunction profile	Spatial shape of the eigenstate.

16 Step 13: Define a Simple Transport Descriptor

For Day 1, we do not compute real conductance yet. Instead, we define a simple proxy:

$$D_{\text{transport}} = \frac{1}{\langle IPR \rangle}. \tag{11}$$

Code:

```
transport_score = 1.0 / np.mean(ipr_values)

print("Transport descriptor:", transport_score)
```

16.1 Physics Meaning

If the mean IPR is high, states are more localized:

$$\langle IPR \rangle \uparrow \quad \Rightarrow \quad D_{\text{transport}} \downarrow . \tag{12}$$

If the mean IPR is low, states are more extended:

$$\langle IPR \rangle \downarrow \quad \Rightarrow \quad D_{\text{transport}} \uparrow . \tag{13}$$

Therefore, this descriptor is not real electrical conductance, but it is a useful first approximation for transport tendency.

16.2 New Terms

Term	Explanation
Transport descriptor	Numerical feature related to electron movement.
Proxy	Approximate measurable quantity.
Conductance	Physical measure of how easily electrons pass through a material.
Delocalization	Spreading of a wavefunction over many sites.

17 Step 14: Generate the First Dataset

Now generate many random defect configurations.

```
def generate_dataset(
    num_samples=200,
    N=40,
    t=1.0,
    epsilon_0=0.0,
    max_defects=10,
    defect_strength_range=(0.5, 4.0),
    seed=42
):
    """
    Generate a dataset of random hydrogen-defect configurations.

    Each sample has:
    - random number of defects
    - random defect strength
    - energy spectrum descriptors
    - localization descriptors
    - simple transport descriptor
    """
    np.random.seed(seed)
    rows = []

    H_clean = build_clean_hamiltonian(N, t, epsilon_0)

    for sample_id in range(num_samples):
        num_defects = np.random.randint(1, max_defects + 1)
        defect_strength = np.random.uniform(*defect_strength_range)

        H_def, defect_sites = add_hydrogen_defects(
            H_clean,
            num_defects=num_defects,
            defect_strength=defect_strength
        )

        eigvals, eigvecs = eigh(H_def)
        ipr = compute_ipr(eigvecs)

        band_gap = eigvals[N // 2] - eigvals[N // 2 - 1]
        mean_ipr = np.mean(ipr)
        max_ipr = np.max(ipr)
        transport_score = 1.0 / mean_ipr

        rows.append({
            "sample_id": sample_id,
            "N": N,
            "num_defects": num_defects,
            "defect_strength": defect_strength,
            "defect_sites": defect_sites.tolist(),
            "mean_energy": np.mean(eigvals),
            "std_energy": np.std(eigvals),
            "min_energy": np.min(eigvals),
            "max_energy": np.max(eigvals),
            "band_gap": band_gap,
            "mean_ipr": mean_ipr,
            "max_ipr": max_ipr,
```

```
            "transport_score": transport_score
        })

    return pd.DataFrame(rows)
```

Run:

```
df = generate_dataset(num_samples=200)
df.head()
```

17.1 Physics Meaning

Each row is one artificial material configuration. Each configuration has:

- a different number of hydrogen defects,
- a different defect strength,
- a different energy spectrum,
- different localization behavior,
- a different transport descriptor.

This becomes the first machine-learning-ready dataset.

17.2 Dataset Columns

Column	**Meaning**
`sample_id`	Index of the generated sample.
`N`	Number of lattice sites.
`num_defects`	Number of hydrogen-induced defects.
`defect_strength`	Local perturbation strength.
`defect_sites`	Lattice positions where defects are placed.
`mean_energy`	Mean of all eigenvalues.
`std_energy`	Standard deviation of eigenvalues.
`min_energy`	Minimum energy eigenvalue.
`max_energy`	Maximum energy eigenvalue.
`band_gap`	Difference between two central energy levels.
`mean_ipr`	Average localization index.
`max_ipr`	Maximum localization index.
`transport_score`	Inverse of mean IPR.

17.3 New Terms

Term	**Explanation**
Sample	One generated defect configuration.
Feature	Input variable for machine learning.
Label	Target class or target value in machine learning.
Descriptor	Physics-based numerical summary.
Dataset	Table of samples and features.

18 Step 15: Save the Dataset

```
df.to_csv("../data/week1_hydrogen_defect_dataset.csv", index=False)

print("Dataset saved to ../data/week1_hydrogen_defect_dataset.csv")
```

Then check summary statistics:

```
df.describe()
```

18.1 Why This Step Matters

Now the first project output exists:

`data/week1_hydrogen_defect_dataset.csv`

This dataset can later be used for:

- classical machine learning regression,
- classical machine learning classification,
- quantum kernel methods,
- Qiskit QSVC,
- later transport modeling.

19 Step 16: Create Dataset Visualizations

19.1 Plot 1: Number of Defects vs Mean IPR

```
plt.figure(figsize=(7, 4))
plt.scatter(df["num_defects"], df["mean_ipr"])
plt.xlabel("Number of hydrogen defects")
plt.ylabel("Mean IPR")
plt.title("Effect of hydrogen defect number on localization")
plt.grid(True)
plt.show()
```

19.2 Plot 2: Defect Strength vs Transport Descriptor

```
plt.figure(figsize=(7, 4))
plt.scatter(df["defect_strength"], df["transport_score"])
plt.xlabel("Defect strength")
plt.ylabel("Transport descriptor")
plt.title("Effect of defect strength on transport descriptor")
plt.grid(True)
plt.show()
```

19.3 Physics Meaning

These plots answer the first scientific questions:

1. Does more hydrogen disorder increase localization?
2. Does stronger defect perturbation reduce the transport descriptor?
3. Are there nonlinear patterns in the generated data?

19.4 New Terms

Term	Explanation
Scatter plot	Plot showing the relation between two variables.
Correlation	Whether two variables change together.
Nonlinear relationship	Relationship that is not a straight line.
Trend	General direction in the data.

20 Step 17: Write the Day 1 Conclusion

At the end of the notebook, write:

> In this notebook, we constructed a simple one-dimensional tight-binding Hamiltonian and introduced hydrogen-induced defects as local on-site energy perturbations. We solved the eigenvalue problem to obtain the energy spectrum and eigenstates. We then computed the inverse participation ratio as a localization index and defined a simple transport descriptor based on the inverse of the mean IPR. Finally, we generated the first dataset of random hydrogen-defect configurations, which can later be used for classical machine learning and quantum machine learning models.

21 Final Notebook Structure

The notebook should contain:

1. Project definition,
2. Imports,
3. Physical parameters,
4. Clean 1D Hamiltonian,
5. Hydrogen defect perturbation,
6. Eigenvalue/eigenvector calculation,
7. Energy spectrum plot,
8. IPR localization index,
9. Most localized eigenstate visualization,
10. Simple transport descriptor,
11. Dataset generation,
12. Dataset visualization,
13. Save dataset,
14. Day 1 conclusion.

22 Most Important Terms for Day 1

Term	Simple Explanation
Lattice	Regular arrangement of sites or atoms.
Hamiltonian	Energy matrix or operator of the system.
Tight-binding model	Model where electrons hop between neighboring lattice sites.
On-site energy	Energy of an electron at a site.
Hopping parameter	Strength of electron movement between neighboring sites.
Defect	Local irregularity in the lattice.
Hydrogen-induced defect	Defect caused or modeled by hydrogen.
Perturbation	Small or local change in a system.
Eigenvalue	Allowed energy level.
Eigenvector	Wavefunction corresponding to an energy.
Energy spectrum	Collection of all energy levels.
Localization	Wavefunction staying in a small region.
IPR	Number measuring how localized a wavefunction is.
Transport descriptor	Approximate feature related to electron movement.
Dataset	Table used for machine learning or quantum machine learning.

23 Day 1 Success Checklist

☐ I created the project folder.

☐ I created the subfolders.

☐ I installed the required packages.

☐ I created `01_lattice_model.ipynb`.

☐ I wrote the project paragraph.

☐ I built the clean Hamiltonian.

☐ I added hydrogen defects.

☐ I computed eigenvalues and eigenvectors.

☐ I computed IPR.

☐ I plotted the energy spectrum.

☐ I plotted localization values.

☐ I visualized the most localized eigenstate.

☐ I created a simple transport descriptor.

☐ I generated a dataset.

☐ I saved the dataset as CSV.

☐ I plotted dataset relationships.

☐ I wrote the Day 1 conclusion.

24 Main Sentence for Today

Today, I built the first physics-informed lattice model and generated the first dataset for hydrogen-defect transport analysis.

25 Mini English Practice

Useful Sentence

I created a tight-binding Hamiltonian and introduced hydrogen defects as local perturbations.

Vocabulary

Word	Meaning	Example Sentence
Perturbation	A small change in a system.	Hydrogen defects are modeled as local perturbations.
Localization	Concentration in a small region.	The IPR measures wavefunction localization.
Descriptor	Numerical feature describing a system.	The transport descriptor summarizes electron spreading.
Hopping	Movement between neighboring sites.	The hopping parameter controls electron motion.
Spectrum	Set of energy levels.	The Hamiltonian gives the energy spectrum.

26 Recommended References

26.1 Tight-Binding and Lattice Hamiltonians

1. David Tong, *Applications of Quantum Mechanics*. Tight-binding discussion.
   https://www.damtp.cam.ac.uk/user/tong/aqm/aqmtwo.pdf

2. Tight-binding model notes.
   https://ssp.utasphys.cloud.edu.au/3-1d/3-3-tightbinding/

3. Kwant tight-binding introduction.
   https://kwant-project.org/doc/latest/tutorial/introduction

26.2 Eigenvalue Calculations

1. SciPy `eigh` documentation.
   https://docs.scipy.org/doc/scipy/reference/generated/scipy.linalg.eigh.html

2. NumPy `eigh` documentation.
   https://numpy.org/doc/stable/reference/generated/numpy.linalg.eigh.html

26.3 Localization and IPR

1. Inverse Participation Ratio overview.
   https://www.emergentmind.com/topics/inverse-participation-ratio-ipr

2. Anderson localization notes.
   https://web.mit.edu/8.334/www/grades/projects/projects19/GuanChenguang.pdf

26.4 Future Quantum Transport

1. Kwant tutorial.
   https://kwant-project.org/doc/dev/tutorial/

26.5 Future Quantum Machine Learning

1. Qiskit Machine Learning quantum kernel tutorial.
   https://qiskit-community.github.io/qiskit-machine-learning/tutorials/03_quantum_kernel.html
2. IBM Quantum kernel training tutorial.
   https://quantum.cloud.ibm.com/docs/tutorials/quantum-kernel-training
3. IBM projected quantum kernels tutorial.
   https://quantum.cloud.ibm.com/docs/tutorials/projected-quantum-kernels

27 Next Step After Day 1

After completing this notebook, the next step is Day 2:

> Extend the defect configurations by varying defect number, defect strength, defect positions, and possibly lattice size. Then compare clean and defected spectra systematically.